What's Next

Future Improvements & Development Roadmap

01 Expanding the Class Library

bullet: *Add new object classes to broaden detection coverage*

The objects we trained on were selected to prove the concept, but the real world contains far more variety. In future iterations, we plan to introduce additional classes — such as glasses, headphones, umbrellas, and other personal items — systematically going through the same rigorous collection, filtering, and labeling pipeline we established. Each new class must meet our 89% accuracy threshold before being considered production-ready. The stronger our class library, the more versatile and commercially applicable the system becomes.

02 Public Testing via an Interactive GUI

bullet: *Launch a user-facing GUI for real people to try the system live*

One of the most honest forms of evaluation is putting your system in the hands of people who had nothing to do with building it. We plan to develop a clean graphical user interface that allows anyone — regardless of technical background — to interact with the model live through their own camera. This removes the controlled lab environment entirely and exposes the model to real diversity: different lighting setups, object orientations, and backgrounds we never anticipated during training.

03 Structured User Feedback Collection

bullet: *Collect structured feedback through a built-in survey form after each session*

Alongside the GUI, we will integrate a simple feedback form that appears after each testing session. Users will be able to rate detection accuracy, flag missed objects, and suggest classes they expected the system to recognize but did not. This transforms every user interaction into a data point for improvement. Rather than guessing what to fix next, we will have direct evidence of where the model falls short in real-world use — a feedback loop that makes every version smarter than the last.

04 Model Optimization for Edge Deployment

bullet: *Compress and optimize the model for mobile and embedded devices*

Currently the model runs on a standard computer with a connected camera. A natural next step is to compress and optimize it for deployment on edge devices — such as Raspberry Pi, mobile phones, or embedded security modules — where processing power is limited. Techniques like model quantization and pruning can drastically reduce model size while preserving most of the accuracy. This opens the door to real-world applications in smart security checkpoints, inventory tracking, and portable detection systems.

05 Multi-Layer Object Identification

bullet: *Move beyond material recognition toward full object identity detection*

Current automated detection systems often stop at recognizing what category an object belongs to. But knowing that something is a "metal object" is not enough — a coin and a key are both metal, yet they require completely different handling in any intelligent system. Our vision for the next phase is a multi-layer identification model that recognizes not just the substance or category, but the specific identity of the object. This deeper level of understanding is what separates a demonstration project from a truly intelligent detection system with real industrial and logistics value.